Large-Scale Mapping of Soil Quality Index in Different Land Uses Using Airborne Hyperspectral Data

Israr Majeed and Bhabani Sankar Das

***Abstract*— Large-scale mapping of soil quality index (SQI) is a challenging task because of the cost and time involved in measuring required soil parameters through conventional wet chemistry-based methods. Hyperspectral remote sensing (HRS) may be used to overcome such a challenge. We hypothesize that soil quality at a specific location may be estimated from remotely sensed reflectance spectra because both these attributes are composite parameters. We used the HRS data collected with the Airborne Visible-Infrared Imaging Spectrometer-Next Generation (AVIRIS-NG) sensor to estimate SQIs in an agricultural catchment. SQIs were developed from 16 different soil properties measured at 101 locations coinciding AVIRIS-NG flight. Chemometric models were used to estimate SQIs from spectral reflectance data collected under laboratory conditions and those processed from AVIRIS-NG data before and after linear and nonlinear unmixing. Except for the linearly unmixed AVIRIS-NG data, three other spectral data sources yielded coefficient of determination (R^2) values exceeding 0.7. Specifically, the R^2 values for the mixed and nonlinearly unmixed spectra were 0.71 and 0.72, respectively, suggesting that HRS approach may directly be used for estimating SQIs. With high validation statistics, we converted the AVIRIS-NG imagery to SQI map for the entire catchment. Such high spatial resolution maps allowed us to examine the effects of land use/cover on soil quality. Strong linear dependencies between SQI and land uses and terrain structures suggested that HRS-derived SQI maps may be used to prioritize soil management efforts for sustainable development.**

***Index Terms*— Airborne Visible-Infrared Imaging Spectrometer-Next Generation (AVIRIS-NG), hyperspectral remote sensing (HRS), land use, nonlinear unmixing, soil quality index (SQI).**

I. Introduction

SOIL quality influences agricultural productivity, plant, and animal health and is essential for sustainable development of our ecosystem [1], [2]. Specifically, soil quality may be used for improving resource use efficiencies across various crop production systems. Typically, a soil quality index (SQI) is estimated by combining selected soil quality indicators [2] based on a minimum data set (MDS) such that the estimated index adequately describes a specific soil function or a relevant ecosystem service [3]. There are various frameworks available for estimating SQIs. Examples include the European Soil Monitoring and Assessment Framework [4], Soil Management Assessment Framework [5], and Cornell Soil Health Test [6], [7], to name a few. Although these frameworks are widely used at different scales (single farms, catchments, and large regions), SQI assessment remains expensive because of the need for analyzing several soil parameters [8]. Consequently, frequent monitoring of soil quality at large scale (e.g., farmers' field and beyond) continues to be a challenging and expensive task.

Over the past few decades, diffuse reflectance spectroscopy (DRS) has emerged as a rapid and noninvasive approach for estimating multiple soil parameters [9], [10]. A few studies have shown that both soil parameters (SQI indicators) and SQIs may be estimated using reflectance spectra collected under controlled laboratory conditions [8], [11], [12], [13]. Specifically, Paz-Kagan et al. [11] employed the DRS approach to estimate 14 SQI indicators and developed a spectral SQI for evaluating soil quality under different land use conditions. Recently, Gozukara et al. [8] combined DRS and portable X-ray fluorescence (pXRF) data to estimate SQIs and other soil indicators along an entire soil profile in a semi-arid environment. With the success of the DRS approach under laboratory conditions, high-resolution spectral data from hyperspectral remote sensing (HRS) platforms are also used for soil quality assessment [14], [15]. The HRS approach provides a contiguous and high spectral resolution reflectance data for estimating soil properties under field scales [16]. Paz-Kagan et al. [14] successfully used the HRS data to estimate SQIs in an experimental farm (total area = 1.62 km^2) in Israel and an agricultural research catchment (total area = 1.44 km^2) in Germany; the HRS approach was shown to have comparable accuracy for estimated SQIs to the ones calculated from laboratory-measured spectral data. Similarly, Levi et al. [15] used the HRS approach to estimate and map both soil properties and SQIs over the Avdat region of Israel (total area = 24 km^2) and concluded that the HRS approach may be used for soil quality assessment under varied environmental conditions. Both the sites from Israel represent desert ecosystem with very limited vegetation; vegetated areas were also masked out from their final analysis. Estimation of SQIs

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through HRS approach for tropical agricultural ecosystems is lacking in soil literature. Specifically, most HRS data analyses adopt a thresholding approach to identify pixels with bare soil conditions [17]. Such an approach precludes the estimation of soil properties for pixels that are not entirely bare. In contrast, an unmixing approach may be adopted for pixels containing multiple endmembers such that soil properties for each pixel in the entire HRS imagery may be estimated to produce a spatially contiguous soil property map [18]. To our knowledge, unmixing of HRS imagery (i.e., without the need for thresholding) has not been used for estimating SQIs for agricultural catchments.

For Indian soils, SQI assessment is in a nascent stage [19] although a few studies have shown SQI estimates based on laboratory-measured soil properties [20], [21], [22]. Studies on SQIs involving conservation agriculture in rice-mustard cropping systems [23] or those for rehabilitating salt-affected soils [24] are reported for Indian soils. A few studies have also used digital soil mapping (DSM) as a data source for estimating SQIs [25]. However, no HRS approach has been used for estimating SQIs in agricultural catchments that contain characteristic left-over stubbles for an extended period of time. With substantial environmental impacts of crop residue burning [26], soil quality assessment in agricultural fields is an important step in the sustainable management of our crop production systems. Thus, the primary objective of this study is to evaluate the potential of airborne HRS data to estimate SQIs in an agricultural catchment. Specifically, we examine whether the HRS data may directly be used for estimating SQI values without the need for any thresholding step.

II. Material and Methods

A. Study Area and Soil Sampling

During 2015–2018, a joint mission between the Indian Space Research Organization (ISRO) and National Aeronautics and Space Administration (NASA) was launched to collect HRS data using the Airborne Visible/Infrared Imaging Spectrometer-Next Generation (AVIRIS-NG) at 157 Indian sites along with ground reference data [27]. As a part of this campaign, HRS image data were collected at two sites (Sites 53 and 54) in the Western Catchment of Chilika lagoon, Odisha, India (see Fig. 1), from 25 December 2015 to 29 December 2015. Concurrent to this campaign, ground truth data consisting of in situ soil spectra, digital photos of sampling locations, and surface soil (0–15 cm) samples were collected at 101 ground reference locations (GRLs). These two sites comprised a total area of about 1932 km^2 within the Western Catchment (19.46°–19.9° N, 85.08°–85.63° E) of the Chilika lagoon. Annual rainfall over the Western Catchment averages at about 1200 mm [28] of which 70% are received during the monsoon period (July–November) leaving most part of the post-monsoon period dry. Typically, December and January are the driest months with the mean annual minimum temperature of 14 °C; the average maximum temperature of 39.9 °C occurs in the summer months. Land use and land cover (LULC) estimates for the Western Catchment suggested that the study area

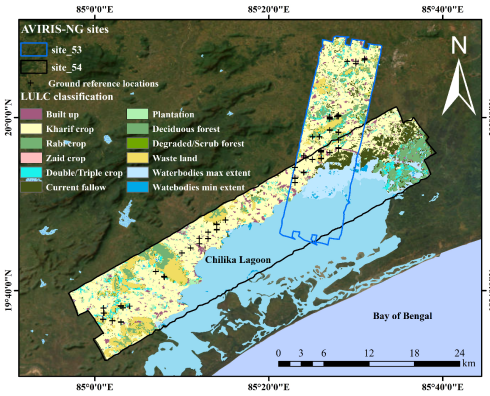


Fig. 1. Sampling locations and the LULC conditions of the study area on the Western Catchment of the Chilika lagoon.

consisted of 531 km^2 under kharif (rainy season) crop, 38 km^2 under rabi (winter season) crop, 27 km^2 under plantation crop, and about 63 km^2 under deciduous forest. Area under two and/or three crops in a year was around 47 km^2. Waterbodies occupied about 355 km^2 area and as much as 469 km^2 were wasteland with 84 km^2 of fallow land. The total built-up area occupied 44 km^2 in the Western Catchment.

Collected soil samples were air-dried, ground, and sifted through a 2-mm sieve and used for physical, chemical, nutrient, and spectral analysis. Each of these samples was analyzed for soil texture, pH, soil organic carbon (SOC) content, cation exchange capacity (CEC), base saturation (BS), and available nutrient contents. Soil texture was analyzed using the pipette method [29]. The chromic acid digestion method was followed to estimate SOC contents [30]. Soil pH and electrical conductivity (EC) were measured in 1:2 and 1:2.5 soil: water slurry, respectively. The Kjeldahl method was used to estimate available N [31]. Available P was estimated using Olsen's and Sommers's [32] method. The ammonium acetate method was followed to estimate the CEC [33]. A flame photometer (Systronics India Ltd., Ahmedabad, India) was used to measure different cations. The diethylenetriamine penta-acetic acid (DTPA) extraction method was used to extract selected micronutrients (Fe, Mn, Cu, Zn, and B) and an inductively coupled plasma (ICP) spectrometer was used for measuring these micronutrients. Gravimetric soil water contents (θ_g) were also measured in each soil sample using oven drying approach.

B. Collection of Airborne and Laboratory Spectra

HRS data consisted of spectral reflectance values over the wavelength range of 380–2510 nm at 5-nm spectra resolution (signal-to-noise ratio: >2000 at 600 nm and >100 at 2200 nm; accuracy: 95%; field of view: 34°; and instantaneous field of view: 1 mrad) resulting in 425 bands per pixel [34], [35]. The AVIRIS-NG sensor was installed on the ISRO B200 aircraft, and multiple flights were conducted at each site (six flights for Site 53 between 10:02 AM to 11:05 AM Indian Standard Time (IST), 26 December 2015; four flights for Site 54 between 10:21 AM and 11:00 AM IST, 27 December 2015). The ground sampling distance (GSD) varied from 4 to 8 m for the flight altitude of 4–8 km, with swaths ranging from 3 to

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6 km [27]. The AVIRIS-NG data products were generated at three levels [27]: level 0 (L0: raw data), level 1 (L1: calibrated radiance), and level 2 (L2: atmospherically corrected surface reflectance). Geometric corrections were applied on L1 and L2 data; atmospherically and geometrically corrected L2 data (spectral reflectance data) were extracted for each sampling location. Atmospheric correction was done using the ATmospheric REmoval (ATREM) algorithm [36]. Out of 425 bands, we used reflectance data only for 321 bands, which had high signal-to-noise ratio. These raw AVIRIS-NG spectra are, hereinafter, referred to as AVIRIS-NG (AVS) spectra. With multiple flights over both the sites, there were overlapping of HRS images both within a site and between two sites. Inspection of overlapping HRS pixels showed that either both the pixels representing a given location contained two valid spectra (coming from two overlapping tiles) or one of the pixels had −9999 (invalid spectra). For example, 24 out of the 101 pixels representing the 101 GRLs had overlapping images; 12 of these had invalid spectra and the remaining 12 pixels had two valid spectra. For the latter scenario, we selected the spectra from the tile that had larger overall reflectance values considering that greater overall reflectance in a specific tile may have resulted from relatively clear sky conditions coinciding AVIRIS-NG flight. We also looked at the images of true color composites and the smoothness of raw AVIRIS-NG spectra for selecting one out of the two valid spectra. We took a such supervisory approach to merge all the ten image tiles such that each image pixel will have a single unique HRS spectrum.

Along with the airborne HRS data, soil spectra were also collected under laboratory conditions using a portable spectroradiometer (Model: Field Spec 3 FR, Analytical Spectral Devices Inc., CO, USA). The contact probe (10-mm spot size) was used to measure spectral reflectance of about 50 g of processed soil samples packed to an aluminum moisture box. The contact probe has an inbuilt halogen light source. The Spectralon white panel (Labsphere, North Sutton, USA) was used as the reference panel during spectral data acquisition. The moisture box was rotated to record four reflectance spectra for every sample. Each soil spectrum was smoothed using a third order Savitzky–Golay filter with a span of 9 nm [37]. Four spectra collected for a given sample were then averaged to obtain the final laboratory soil spectra (LSS) for that sample for chemometric analyses.

C. Estimation of SQIs

A total of 16 soil parameters (sand content, clay content, pH, EC, SOC, CEC, BS, θ_g, and plant-available forms of N, K, P, Fe, Mn, Cu, Zn, and B) were measured in each soil sample for selecting key SQI indicators following the principal component analysis (PCA) approach. To minimize the number of components, the eigenvalue criterion was utilized [38]; components with eigenvalue ≥ 1 were selected to constitute the MDS [39]. Component loadings assigned to variables under a specific principal component (PC) were utilized to reduce the number of variables. Under each PC, variables within 10% of the highest component loading were selected [40].

TABLE I
CLASSIFICATION OF SQI IN RELATION TO PLANT GROWTH PROPOSED BY QI ET AL. [46]

SQI Grade	SQI Range	SQI Rating	Description
I	>0.85	Very high	Most suitable for plant growth
II	0.7-0.85	High	Suitable for plant growth
III	0.55-0.7	Moderate	Suitable for plant growth with some limitations
IV	0.40-0.55	Low	More limitations than grade III
V	<0.4	Very low	Most severe limitations for plant growth

Pearson correlation coefficients (r) were estimated to remove redundant indicator variables. Among well-correlated variables (i.e., variables with $r > 0.70$), only the variable with the highest factor loading (absolute value) was kept in the MDS [41].

Generally, two scoring methods (linear and nonlinear) and three integration methods (additive, weighted additive, and Nemoro) are followed to estimate SQI [42], [43]. We estimated all the six different SQIs and performed sensitivity analysis [44] to identify the best SQI for our study area. Results of the sensitivity analysis showed that the linear scoring and weighted additive integration methods yield the best SQI for our study site (data not shown). Therefore, all the selected parameters in MDS were transformed to unit less scores between 0 and 1 using the linear scoring method (more is better, less is better, and optimum is better). In the case of "more is better" function, the linear score (S_{lin}) was calculated by dividing each indicator variable (X) with its maximum value ($X_{\max}$). For the "less is better" function, S_{lin} was estimated by taking the ratio between the minimum value of the indicator variable ($X_{\min}$) and X. For the "optimum is better" function, S_{lin} was estimated by combining both these cases

$$S_{\text{lin}} = \begin{cases} X/X_{\max}, & X < X_{\text{opt}} \\ X_{\min}/X, & X > X_{\text{opt}} \end{cases} \tag{1}$$

where X_{opt} denotes the optimum threshold value for X. Scoring functions for respective variables are listed in Table S1 in the Supplementary Material. These dimensionless scores are then linearly combined to obtain the weighted additive SQI

$$\text{SQI} = \sum_{k=1}^{K} w_k \cdot S_k \tag{2}$$

where w_k and S_k are the kth weight and score of the individual parameter, respectively, and K is the number of soil indicators in the MDS [45]. We further classified soil quality into five grades (see Table I) based on the SQI rating criteria proposed by Doran and Parkin [46].

D. Unmixing of AVIRIS-NG Data

Ground observations at 101 GRLs suggested that the study area has a heterogeneous mix of bare soil, non-photosynthetic vegetation (NPV), and green photosynthetic vegetation (GPV) (see Fig. 2). Consequently, we applied the linear polynomial mixing (LPM) approach [18], [47] to unmix each raw AVS

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spectrum into respective endmember spectra. Specifically, the linear extended algorithm [48] was utilized to unmix the AVS spectra. We also used the linear mixing (LM) algorithm [49] for unmixing AVS spectra. In the LPM approach, the AVS spectra of a given pixel are expressed as the sum of both linear and polynomial mixture terms [18], [47]

$$\begin{aligned} x_i(n) = {} & a_1(i)s_1(n) + a_2(i)s_2(n) + a_3(i)s_3(n) \\ & + a_{1,2}(i)s_1(n)s_2(n) + a_{1,3}(i)s_1(n)s_3(n) \\ & + a_{2,3}(i)s_2(n)s_3(n) + a_{1,2,3}(i)s_1(n)s_2(n)s_3(n). \end{aligned} \quad (3)$$

The LPM model extends the three-endmember LM model, treating each AVIRIS-NG pixel as a linear-polynomial mixture of various endmembers. Equation (3) can be expressed in a matrix form to accommodate nonnegative matrix factorization (NMF) methods [48]

$$X = AS = A_a S_a + A_b S_b + A_c S_c \quad (4)$$

where the matrix X represents the AVS spectra, A is the mixing coefficient matrix, and S is the source matrix of endmember spectra. The matrices A and S may be given as

$$A = [A_a A_b A_c]$$

$$S = \begin{bmatrix} S_a \\ S_b \\ S_c \end{bmatrix}$$

with A_a, A_b, and A_c defined as

$$A_a = \begin{bmatrix} a_1(1) & a_2(1) & a_{3,}(1) \\ \vdots & \vdots & \vdots \\ a_1(P) & a_2(P) & a_{3,}(P) \end{bmatrix}$$

$$A_b = \begin{bmatrix} a_{1,2}(1) & a_{1,3}(1) & a_{2,3}(1) \\ \vdots & \vdots & \vdots \\ a_{1,2}(P) & a_{1,3}(P) & a_{2,3}(P) \end{bmatrix}$$

$$A_c = \begin{bmatrix} a_{1,2,3}(1) \\ \vdots \\ a_{1,2,3}(P) \end{bmatrix}.$$

Similarly, the matrices S_a, S_b, and S_c may be defined as

$$S_a = [s_1 \quad s_2 \quad s_3]^T$$

$$S_b = [s_1 \odot s_2 \quad s_1 \odot s_3 \quad s_2 \odot s_3]^T$$

$$S_c = [s_1 \odot s_2 \odot s_3]^T.$$

Following the recommendation of Meganem et al. [48], we adopted the classical Frobenius norm as the criterion for minimizing errors for solving (4):

$$J = \frac{1}{2}\|X - \mathrm{AS}\|_F^2. \quad (5)$$

The algorithm also aims to meet the criterion that the fractions are nonnegative fractions and will sum to one within a pixel. Soil spectra obtained through linear and nonlinear unmixing are hereinafter, referred to as AVS_L and $\mathrm{AVS}_{\mathrm{NL}}$, respectively. The pseudocode [18] for the LPM algorithm is given in Algorithm 1.

Algorithm 1 requires reference spectra and initial abundance values for three selected endmembers (bare soil, GPV, and NPV) to initiate unmixing. Laboratory-measured soil spectra and dried crop residue spectra were averaged to obtain the reference spectra for bare soil and NPV, respectively. Because no laboratory-measured GPV spectra were available, we selected ten locations based on dense vegetation as revealed in the true color composite of the AVIRIS-NG image. Corresponding AVS spectra of these locations were then averaged to obtain the reference GPV spectra. Similarly, the linear unmixing algorithm [49] integrated in the recently developed MATLAB Hyperspectral Image Processing Toolbox (The MathWorks Inc., USA) was used to estimate initial abundance values for each endmember.

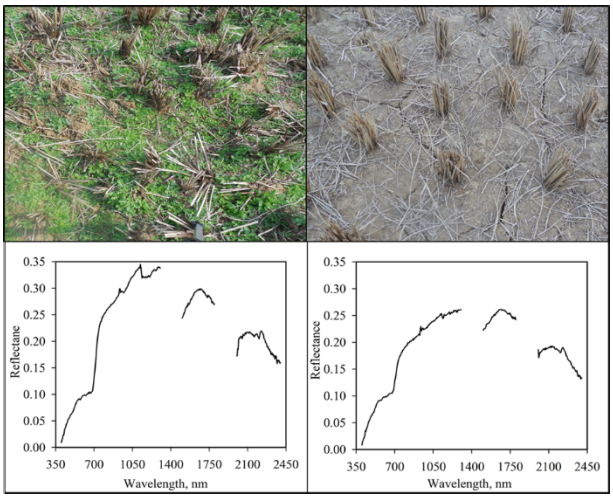


Fig. 2. Photographs showing three potential endmembers (bare soil, GPV, and NPV) for two randomly selected GRLs and their corresponding AVIRIS-NG spectra (AVS).

E. Preprocessing of Spectra and Chemometric Modeling

Before running chemometric models, first derivative [50] and the standard normal variates [51] were estimated from each spectrum as a spectral preprocessing step. The details of preprocessing techniques are given in Supplementary Materials. Two different chemometric models (PLSR: partial-least-squares regression and $\mathrm{PLSR}_{\mathrm{FS}}$: feature selection-based PLSR) were calibrated to estimate SQI values using the LSS, AVS, AVS_L, and $\mathrm{AVS}_{\mathrm{NL}}$ spectral sources. The PLSR method maximizes the covariance between features in the projected space and the response variables [52]. For the $\mathrm{PLSR}_{\mathrm{FS}}$ approach, soil properties are estimated in the PLSR approach after selecting important feature variables out of a set of 35 predictor variables consisting of PLSR regression coefficients (β), variable influence on projection (VIP), mutual information based adjacency vector (AMI) [53], and their combinations [54]. The ordered predictor selection (OPS) approach is used to identify a parsimonious set of predictors in this algorithm. We also used the partition sorting approach [55] to divide our whole dataset into 75% calibration and 25% validation samples. Agreement between measured (Y_i) and predicted ($\hat{Y}_i$) soil properties was assessed using the root-mean-squared error (RMSE) and the coefficient of determination (R^2)

$$\mathrm{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}\left(Y_i - \hat{Y}_i\right)^2} \quad (6)$$

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Algorithm 1 Linear Polynomial Mixture (LPM) Model

for $i \leftarrow 1$ to P**do**

for $t \leftarrow 1$ to maximum number of iterations **do**

1: *Update the source matrix S:*

$$S \leftarrow S \odot ((A^T X)/.(A^T AS + \varepsilon)).$$

2: *Update the mixing matrix A:*

$$A \leftarrow A \odot \left(\left(XS^T\right)/.\left(ASS^T + \varepsilon\right)\right)$$

3: *Constraints on matrix A is given as:*

$$[a_i(i) \ldots a_M(i)] \leftarrow [a_i(i) \ldots a_M(i)]/\sum_{j=1}^{M} a_j(i)$$

$$a_{j,k}(i) \leftarrow \min\left\{a_{j,k}(i), 0.5\right\}$$

$$a_{j,k,l}(i) \leftarrow \min\left\{a_{j,k,l}(i), 0.5\right\}$$

4: *Error estimation:*

$$\widehat{X} \leftarrow AS$$

$$Err_{tot} \leftarrow \frac{\left\|X - \widehat{X}\right\|_F}{\|X\|_F}$$

end for

5: *Pure spectra matrix* $\widehat{S_n}$ *:*

Estimate $\widehat{S_n}$ from the updated A and S

$$\widehat{S_n} \leftarrow A_a S_a$$

end for

P: number of pixels; M: the number of sources present in the image; $a_j(i)$, $a_{j,k}$, (i), and $a_{j,k,l}$ (i) are the mixing coefficients corresponding to linear, quadratic and polynomial parts of the model, respectively; Err_{tot}: total error; The ε term in the denominator is a small and positive value added to avoid division by zero. The notations $\odot$ and /. correspond to element-wise multiplication and division, respectively.

$$R^2 = 1 - \frac{\sum_{i=1}^{n}\left(Y_i - \hat{Y}_i\right)^2}{\sum_{i=1}^{n}\left(Y_i - \bar{Y}\right)^2} \tag{7}$$

where $\bar{Y}$ is the average value of Y_i and n is the number of GRLs.

III. Results and Discussion

A. Descriptive Statistics of Soil Quality Indicators

Table II lists the descriptive statistics for soil physical and chemical properties and selected nutrient contents measured in collected soil samples from 101 GRLs. In general, soils from the study area showed a wide range of soil textural fractions with the average sand and clay contents of 59% and 24%, respectively; sandy clay loam was found to be the dominant textural class. Based on soil fertility classification ratings [56], collected soil samples were moderately acidic (average pH = 6.28) although some of the soil samples also showed moderately alkaline reaction (pH range: 4.7–8.94). Chilika lagoon shows a gradient of salinity being connected to the Bay of Bengal on the east through a mouth [57], which may be the reason for high soil pH in some of the GRL samples. The average SOC content of collected samples was 0.83% (coefficient of variation (CV) = 40%). With the hillslope hydrology of the study area and presence of deciduous forests on the western part of the catchment [58], almost 50% of the samples showed relatively high SOC contents possibly because of the deposition of organic matter (forest litter) in low-lying areas. In general, available N, P, and K contents were low with their average values of 151, 7.8, and 200 kg·ha^{-1}, respectively, indicating possible deficiency of all the three major nutrient elements in these soils. The CEC values for collected soil samples ranged from 3.3 to 52.2 cmol$_c$·kg^{-1} (mean = 17.3 cmol$_c$·kg^{-1}; CV = 76%). High CEC values in a few samples may be because of the greater proportion of illitic clays although soils in this region are primarily kaolinitic in nature. Vasava et al. [59] used X-ray diffraction measurements in selected soil samples collected during a previous sampling campaign from the same area and observed the presence of both kaolinitic and illitic clays in their samples. The average values for the available Fe, Mn, and Cu contents in collected soil samples were 48.08, 65.97, and 2.80 mg·kg^{-1}, respectively, indicating the presence of high levels of these micronutrients in the study area. Available B content was also low (mean = 0.43 mg·kg^{-1}) although available Zn content was generally sufficient (mean = 0.95 mg·kg^{-1}). Table II also shows that most of the collected soil samples were relatively dry at the time of sampling with the average θ_g value of 14%; 76 out of 101 soil samples had θ_g values less than 20%.

B. Minimum Dataset and SQI

Our goal was to create a comprehensive SQI that encompasses physical, chemical, and biological attributes of soils while keeping it closely aligned with agriculture. Therefore, we included all the basic soil properties (texture, SOC, pH, and EC) and considered SOC as a good proxy for the biological characteristics of our soils. Both macronutrients and micronutrients were also included because of their key roles in rapidly increasing agricultural intensification in our country. Specifically, Indian soils are already showing deficiency symptoms for micronutrients such as Fe, Zn, S, B, and Cu in addition to major nutrient elements of N, P, and K [19]. With no hydraulic properties available for our study site, we also included soil wetness in our overall dataset. With these 16 measured soil properties at 101 GRLs, the PCA results revealed that five PCs had eigenvalues ≥ 1 and accounted for 73% of the total variability in the dataset (see Table III). The component loading matrix revealed that sand, clay, CEC, and available B are the highly weighted variables accounting for 31% of the total variance under PC1. Based on the correlation analysis (see Fig. 3), clay content and CEC were selected for the MDS. The PC2 (20% of variance) revealed that available Fe, Cu, and Zn are the highly weighted variables. With $r > 0.7$ among these three variables (see Fig. 3), available Fe was selected for the MDS because of its highest factor loading. For PC3 (8% of variance), available N was the only highly weighted variable and was retained in the MDS. For PC4 (8% of variance) and PC5 (8% of variance), and EC and available P were the highly weighted variables, respectively, and were included in the MDS. Thus, the MDS consisted of clay content, CEC, available Fe, available N, EC, and available P. Corresponding

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TABLE II

DESCRIPTIVE STATISTICS FOR SOIL PHYSICAL AND CHEMICAL PROPERTIES AND NUTRIENT CONTENTS MEASURED IN SOIL SAMPLES COLLECTED FROM 101 GRLS ACROSS THE WESTERN CATCHMENT OF THE CHILIKA LAGOON

Properties	Minimum	Maximum	Mean	CV	Skewness	Kurtosis
Sand content, %	0.2	85	59	29	-0.6	0.4
Clay content, %	5	55	24	42	0.8	0.2
pH	4.70	8.94	6.28	15	0.59	-0.05
EC, μS cm^{-1}	33	854	169	77	2.0	8.0
SOC content, %	0.21	2.08	0.83	40	0.75	1.49
CEC, $cmol_c$ kg^{-1}	3.3	52.2	17.3	76	1.6	2.0
Base saturation, %	22	95	63	53	3.85	23.22
Av. N	44	314	151	23	1.0	5.0
Av. K	85	429	200	40	0.9	0.3
Av. P	0.06	97	7.8	148	5.46	37.80
Av. Fe	2.3	171.1	48.1	82	1.00	0.39
Av. Mn	5.3	140.5	66.0	58	0.26	-0.95
Av. Cu	0.60	7.09	2.80	58	0.73	-0.24
Av. Zn	0.22	3.55	0.95	63	1.49	2.99
Av. B	0.10	1.90	0.43	75	1.95	4.59
θg, g g^{-1}	0.01	0.43	0.14	71	0.90	-0.05

CV: Coefficient of variation expressed in percentage; SOC: soil organic carbon; CEC: cation exchange capacity; Av.: available; θ_g: gravimetric water content; macronutrients are in kg ha^{-1} and micronutrients are in mg kg^{-1}

weights (see Table III) and these indicators were combined using (2) to yield an expression for the SQ

$$\mathrm{SQI} = 0.422\left(S_{\mathrm{clay}} + S_{\mathrm{CEC}}\right) + 0.272S_{\mathrm{Fe}} + 0.115S_N + 0.105S_{\mathrm{EC}} + 0.086S_P. \quad (8)$$

Equation (8) suggests that soil clays and CEC have the largest contribution (∼42%) to SQI followed by available Fe contents. Based on factor loading (see Table III), we found that both clay contents and CEC have to be selected as indicators for constructing SQI; hence, both these indicators are assigned the same weights. While clay contents represent the amount of clay present in a soil sample, CEC is an indirect measure of the nature of soil clays (expanding versus nonexpanding clays) and the amount of SOC present in a soil sample. Nonexpanding clays such as kaolinites have characteristically low CEC values (in the range of 5–20 $cmol_c \cdot kg^{-1}$) while expanding clays (montmorillonite) and SOC contents have more than five times CEC values than kaolinitic clays. Because CEC is a measure of nutrient retention capacity of soils and clay contents significantly influence both water retention characteristics and soil structure, these two soil properties should naturally be major components of soil quality. Indian soils are typically low in SOC contents (∼0.54% [60]); for our study site, the average SOC was 0.83% with a positive skewness coefficient of 0.75. A narrow range of SOC may be a reason behind not having SOC as a key SQI indicator although its closely linked available N was selected as an indicator. Thus, SOC was implicitly selected as the SQI indicator in our dataset. The selection of Fe as an indicator stems from granitic parent material and prevailing high weathering intensities over our study site [61], [62]. High Fe content is a characteristic feature of eastern Indian coastal soils [63] and our analysis rightly captured Fe as a significant SQI indicator. Finally, because most of the soil samples were collected from

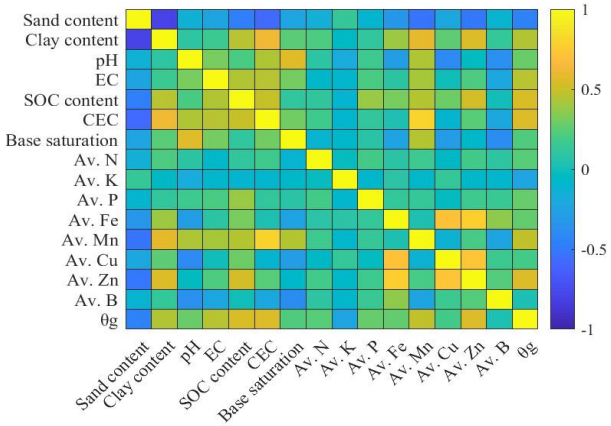


Fig. 3. Heat map showing Pearson's correlation coefficient between different soil properties.

the agricultural fields, available N, EC, and available P are significant parameters contributing to SQI.

Estimated SQI values ranged from 0.19 to 0.62 with the average value of 0.40. Analysis of soil quality grades (see Table I) suggested that 50% of our GRLs had very low soil quality (grade V), 44% had low soil quality (grade IV), and the remaining had moderate soil quality (grade III). With generally low CEC and available major nutrients (N and P), most soil samples showed poor soil quality grades for the study area. Soil samples classified as grade IV were commonly observed in areas where kharif crop is the predominant land use, indicating comparatively better soil quality in these regions compared to other land uses. On the other hand, the majority of samples falling under grade V were found in areas classified as wastelands and double/triple crop. Samples collected from these wastelands exhibited low CEC and limited availability of nutrients, which eventually reflected in having these soils classified under soil quality grade V. Lack of sufficient land

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TABLE III
PC ANALYSIS RESULTS AND FACTOR LOADINGS FOR SOIL QUALITY INDICATORS

Principal components	PC1	PC2	PC3	PC4	PC5
Eigen values	4.96	3.20	1.35	1.24	1.01
% Variance	31	20	8	8	6
% Cumulative variance	31	51	59	67	73
Weights	0.422	0.272	0.115	0.105	0.086
Indicators	Factor loading*				
Sand content, %	0.815	0.271	0.037	0.062	0.169
Clay content, %	**0.861**	0.312	0.116	0.025	0.055
pH	0.319	0.531	0.054	0.495	0.029
EC, µS cm^{-1}	0.212	0.006	0.240	**0.738**	0.032
SOC content, %	0.379	0.328	0.114	0.633	0.029
CEC, $cmol_c\ kg^{-1}$	**0.803**	0.128	0.046	0.343	0.057
Base saturation, %	0.407	0.437	0.335	0.332	0038
Av. N	0.241	0.049	**0.809**	0.000	0.080
Av. K	0.107	0.040	0.524	0.623	0.124
Av. P	0.091	0.135	0.075	0.143	**0.896**
Av. Fe	0.014	**0.896**	0.141	0.038	0.116
Av. Mn	0.094	0.422	0.416	0.261	0.467
Av. Cu	0.362	0.785	0.072	0.297	0.089
Av. Zn	0.142	0.864	0.078	0.105	0.005
Av. B	0.787	0.186	0.025	0.341	0.091
θg, g g^{-1}	0.468	0.221	0.123	0.580	0.195

SOC: soil organic carbon; CEC: cation exchange capacity; Av.: available; θg: gravimetric water content; macronutrients are in kg ha^{-1} and micronutrients are in mg kg^{-1}; *bold face underlined factor loadings indicate selected soil properties retained in the minimum dataset (MDS)

TABLE IV
MEAN AND STANDARD DEVIATION (σ) OF SAM FOR SOIL SPECTRA EXTRACTED USING LINEAR AND NONLINEAR UNMIXING APPROACHES AT 101 GRLS

Parameter	Bare soil	
Method	Linear unmixing	Non-linear unmixing
Mean	0.189	0.186
σ	0.135	0.135

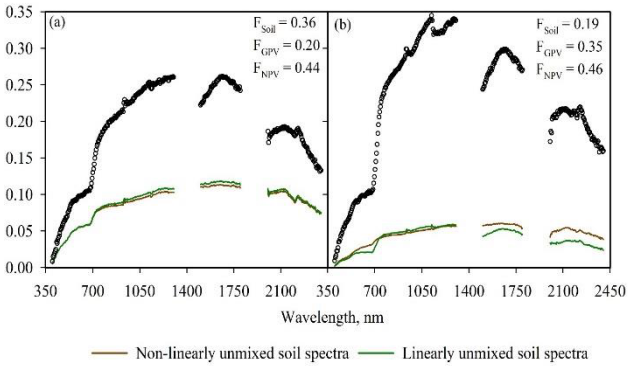


Fig. 4. Raw AVIRIS-NG spectra (circles) and soil spectra extracted using linear (green line) and nonlinear (brown line) unmixing approaches for selected locations with (a) low and (b) high green vegetation fractions.

management practices may have led to soil erosion and nutrient depletion causing overall soil degradation in these areas.

C. Performance of Unmixing Algorithm

We calculated the spectral angle mapper (SAM) values to evaluate the accuracy of linear and nonlinear unmixing approaches for estimating soil spectra in each pixel. Table IV lists the mean SAM values for the unmixed soil spectra estimated using linear unmixing ($\mu = 0.189$) and nonlinear unmixing ($\mu = 0.186$). These values indicate high spectral similarity between the estimated and reference soil spectra. Fig. 4 shows the observed AVS spectra and unmixed soil spectra estimated using both linear and nonlinear approaches for two randomly chosen GRLs having different fractional abundances. Both linear and nonlinear unmixing approaches could clearly capture the characteristic Si–OH absorption feature typically observed at 2200 nm in a soil spectrum because of metal–OH stretching [64], [65]. Fig. 4 also shows that both linear and nonlinear unmixing methods yield similar spectra for pixels containing more bare soil fraction [$F_{soil} = 0.36$ in Fig. 4(a)]. However, when the bare soil fraction in a pixel is low [$F_{soil} = 0.19$ in Fig. 4(b)], soil spectra obtained with linear (green line) and nonlinear (brown line) unmixing had different shapes over the visible part of the spectrum. Specifically, soil spectra obtained through linear unmixing show the characteristic absorption peak and prominent red edge typically seen with photosynthetic vegetation, which should not be present in a soil spectrum. Spectra obtained from nonlinear unmixing rightly do not show any such vegetation-specific feature suggesting that nonlinear unmixing should be followed in our dataset.

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TABLE V

COEFFICIENT OF DETERMINATION (R^2), RMSE, AND RATIO OF PERFORMANCE TO INTERQUARTILE DISTANCE (RPIQ) FOR SQI VALUES ESTIMATED USING DIFFERENT SPECTRAL DATA SOURCES; DATA ARE PRESENTED FOR THE BEST SPECTRAL TRANSFORMATIONS AND BEST MODELS

Spectral data sources	R^2	RMSE	RPIQ	Best model
LSS	0.85	0.04	4.39	PLSR
AVS	0.71	0.06	3.07	$PLSR_{FS}$
AVS_L	0.53	0.08	1.41	$PLSR_{FS}$
AVS_{NL}	0.72	0.06	3.11	$PLSR_{FS}$

LSS: laboratory soil spectra; AVS: AVIRIS-NG spectra; AVS_L: soil spectra obtained with linear unmixing; AVS_{NL}: soil spectra obtained with nonlinear unmixing; PLSR: partial-least-square regression; $PLSR_{FS}$: feature selection-based PLSR

D. Estimation and Mapping of SQIs Using HRS Data

Table V shows the performance statistics of selected chemometric models for estimating SQIs using LSS, AVS, AVS_L, and AVS_{NL}. The best-performing model was considered to be the one with the highest R^2 (and lowest RMSE) value across two spectral transformations. Although differences in prediction accuracies may be observed using LSS, AVS, AVS_L, and AVS_{NL}, all the spectral data sources predicted SQI with reasonably good accuracy except for AVS_L. As expected, soil spectra collected under the controlled laboratory conditions (LSS spectra) showed the highest R^2 of 0.85 (and the lowest RMSE value of 0.04) in the validation dataset (see Table V). Similar accuracies were obtained by Gozukara et al. [8], Askari et al. [12], and Levi et al. [13] who used the DRS approach to estimate SQIs using spectral data collected under laboratory conditions and soils from different land management systems. Askari et al. [12] reported an R^2 value of 0.89 (RMSE = 0.03) for grassland and an R^2 value of 0.84 (RMSE = 0.05) for arable land. Levi et al. [13] reported an R^2 value of 0.90 (RMSE = 0.06) in cross-validated data obtained from dryland areas, while Gozukara et al. [8] obtained an R^2 value of 0.93 using the DRS approach.

For the AVIRIS-NG data, the AVS and AVS_{NL} resulted in better performance for the DRS approach than the linear unmixing of AVS. Fig. 5 shows the observed versus predicted SQI values obtained using the best modeling approach and best spectral transformations for the specific spectral data source. Observed and predicted SQIs values are closely distributed along the 1:1 line suggesting that the LSS and AVS, and AVS_{NL} data sources may be combined with the chemometric models for estimating SQI values. Table V shows that the validation dataset yielded R^2 values of 0.71 and 0.72 when the mixed AVS spectra and the AVS_{NL} were used as the spectral data source, respectively. Low prediction accuracy obtained with AVS_L (R^2 = 0.53), on the other hand, indicates that linear unmixing is not adequately suited for unmixing AVS when endmembers are present as heterogeneous mixtures in a pixel. The presence of vegetation features in soil spectra in linearly unmixed AVIRIS-NG spectra may be the reason

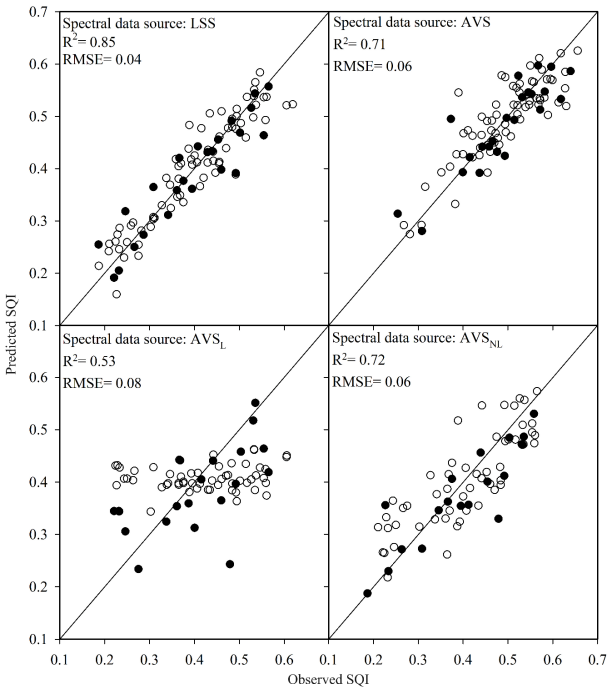


Fig. 5. Observed and predicted SQI values obtained using different spectral data sources. LSS: laboratory soil spectra. AVS: AVIRIS-NG spectra. AVS_L: soil spectra obtained with linear unmixing. AVS_{NL}: soil spectra obtained with nonlinear unmixing. Unfilled and filled circles represent calibration and validation data, respectively.

behind such a result. Keshava and Mustard [49] suggested that linear unmixing is not adequately suited for unmixing AVS when multiple endmembers are present as heterogeneous mixtures in a pixel. This suggests that the traditional linear unmixing approach may fail to accurately separate respective endmember spectra from a single pixel. No significant difference in the prediction accuracies between AVS and AVS_{NL} may be a key result in our study, which suggests that unmixing may not be required in the first place when estimating composite soil parameters such as SQI from raw hyperspectral data. We consider this as a key result because no elaborate preprocessing step (such as unmixing) may be needed for estimating SQI from HRS data. The reason behind such a result is that SQI is a composite attribute of soil present in a pixel. Similarly, raw HRS spectrum is a measure of the overall reflectivity of that pixel. Strong linkages between two composite (averaged) attributes may be the reason behind better performance statistics of the chemometric model when raw HRS spectra are used for estimating SQI. Our finding has an important implication for hyperspectral data analysis: applications that rely on estimating composite indices such as SQI may not need unmixing of hyperspectral data. This would save computational burden and avoid potential errors introduced by unmixing algorithms. Such a general result should be verified with other composite soil indices (e.g., soil fertility index).

The best prediction model built with the AVS_{NL} data source ($PLSR_{FS}$) collected from 101 GRLs was applied to each pixel of AVIRIS-NG scenes to generate an SQI map (see Fig. 6) for

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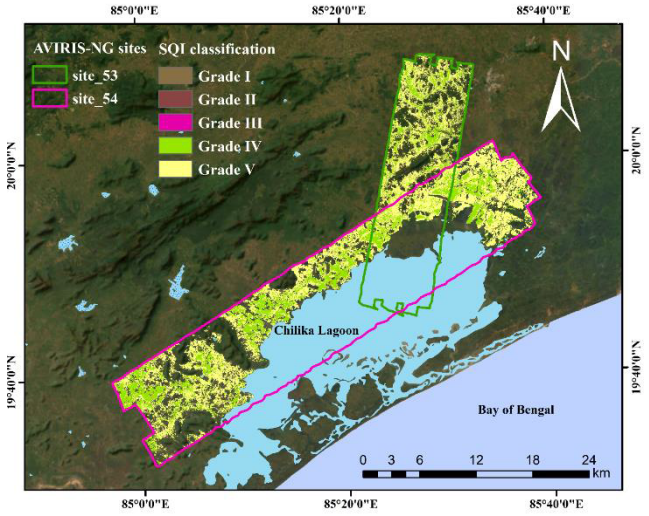


Fig. 6. SQI map developed using the AVIRIS-NG data across the study area using the best modeling approach.

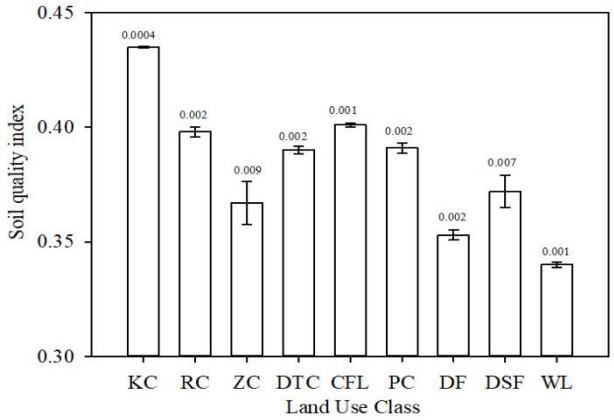


Fig. 7. SQI under different land use systems. KC: kharif crop. RC: rabi crop. ZC: zaid crop. DTC: double/triple crop. PC: plantation crop. DF: deciduous forest. WL: waste land. CFL: current fallow land. DSF: degraded/scrub forest. The error bar and the annotated value show the respective standard error of mean for the specific land use class.

the entire study site. The generated SQI map was classified into five grades (see Table I) to identify different grades of soil quality in the study area. Overall, the soil quality of the study area was mainly grades IV and V (more limitations than grade III; low grade) covering 37% and 61% of the study area, respectively. The remaining area accounted for grade III soil quality. The classification of study area into different grades of soil quality may help identify areas with limitations for crop growth. This may provide insights into the areas where investments in soil quality would be most effective. These maps may provide valuable information to managers and farmers to implement targeted management practices to improve soil quality and maximize crop productivity. Thus, SQI maps generated with HRS data may provide a spatially continuous measure of soil quality across a large agricultural landscape. Overall, the use of HRS data and SQI maps may be an effective tool for sustainable soil management and agricultural planning. In particular, the HRS approach may aid in the mapping of inaccessible areas by minimizing the need for time-consuming and expensive field surveys.

E. Evaluation of Soil Quality Under Different Land Use

With the SQI mapping of the entire study area, we examined how different land uses are linked with the prevailing soil quality. Fig. 7 shows the average SQI values and their standard errors for nine different land use types (e.g., kharif crop, current fallow, rabi crop, plantation crop, double/triple crop, deciduous fine, zaid crop, degraded/scrub forest, and wasteland) present in our study site. Although areas under kharif crop show the highest SQI values among the listed land use types, almost all the land uses show low soil quality grade (SQI $<$ 0.55). We also performed the analysis of variance (ANOVA) to test the level of significance in SQI differences across various land use classes (see Tables S2 and S3). Results indicate that the mean differences in SQI values between most land uses are statistically significant (p-value $<$ 0.05). Specifically, the SQI values for the areas under kharif crop were significantly larger than other land use classes with differences ranging from 3% for the fallow land to about 10% for the areas under wasteland (see Table S3). This suggests that farmers prefer to grow kharif crops wherever relatively higher quality soils are available. Double/triple cropping involves growing multiple crops in a single year, exerting high demands on soil nutrients, and leading to potential nutrient imbalances and degradation of soil quality over time. Fig. 7 also shows that areas under waste land and degraded/scrub forest exhibited the lowest soil quality in the study area. Waste lands, which may have been abandoned or degraded due to human activities, often lack essential nutrients and organic matter. Similarly, degraded or scrub forests may have experienced soil erosion and nutrient depletion resulting in degraded soil quality. Areas under zaid crop showed significantly lower SQI values than those under rabi and kharif crops. However, no significant difference was found between SQI values under zaid crop and other land use classes except for the wasteland (see Table S3).

With different cropping patterns, deciduous forests, and waste lands covering a major portion of the study area, we linked the average SQI values for each existing land use with selected terrain attributes with the assumption that terrain structure will strongly influence soil quality under hillslope hydrologic conditions. Fig. 8 depicts the relationship between the land use-specific SQI values with elevation, slope, terrain ruggedness index (TRI), and topographic wetness index (TWI). To begin with, we analyzed Pearson correlation (r) between SQI and 15 different terrain indices. Out of these, these four attributes (see Fig. 8) showed r values exceeding 0.7. Negative r values for elevation, slope, and TRI suggest that SQI values tend to decrease with the increase in these three terrain parameters. A positive correlation between SQI and TWI suggests that soil wetness has a favorable influence on SQI. In summary, higher elevations, steeper slopes, and more rugged terrains are associated with lower soil quality, while an increase in topographic wetness is linked to better soil quality.

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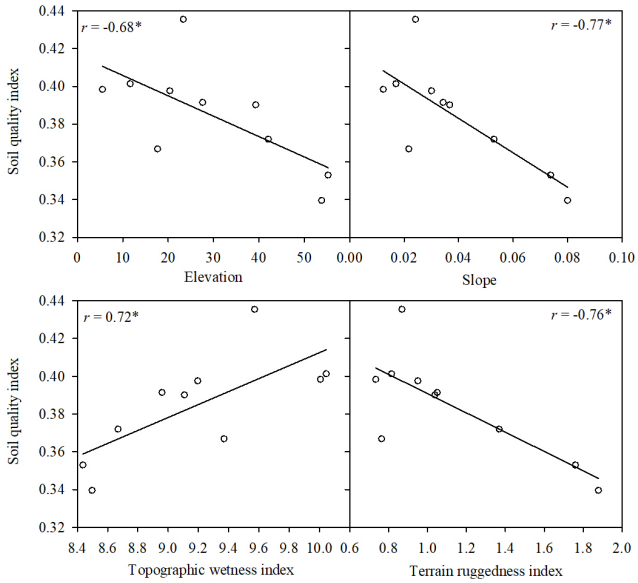


Fig. 8. Relationship between average SQI and terrain attributes for different land uses. r: Pearson correlation coefficients (r) values; *significant at $p < 0.05$.

To improve soil quality, it is essential to assess and address the specific factors contributing to poor soil quality. Implementing sustainable land management practices, such as conservation agriculture, soil erosion control measures, and proper nutrient and water management, may restore and enhance soil quality over time. The SQI mapping results provide valuable information for policymakers and land managers to make informed decisions regarding land use planning and management. They can use this information to identify areas that require additional interventions to improve soil quality and promote sustainable land management practices.

IV. Conclusion

Soil quality assessment in agricultural fields is commonly done by estimating SQIs using laboratory-measured soil properties. Here, we examined the potential of hyperspectral imaging data as a comprehensive, cost- and time-efficient solution for monitoring soil quality in a 1932-km^2 agricultural catchment. In addition to the airborne AVIRIS-NG hyperspectral data, we also measured LSS for 101 soil samples collected from this catchment for estimating SQIs. We combined different scoring methods and weighted averaging to estimate SQI values from key soil parameters. Two unmixing approaches (linear and nonlinear) were used to unmix AVIRIS-NG data. For developing a comprehensive chemometric model for SQIs, we compared two different spectral transformations and two different machine learning models using both proximal and remotely sensed spectral data sources. Among 16 different soil properties considered, the PCA showed that clay content, CEC, available Fe, N, and EC and K contents are the key indicator variables for soil quality in our catchment. With the available N selected as a key quality parameter, future studies in agricultural fields should include specific biological characteristics while designing SQIs. Results show that SQI could be predicted with a reasonable accuracy ($R^2 > 0.7$) for all the spectral sources. Results show that no significant difference is observed when comparing the prediction accuracies between AVS and AVS_{NL}. High prediction accuracy of AVS data showed that continuous maps for SQIs may be conveniently developed for different land use systems. High-resolution SQI maps for such large spatial scales may help identify areas with limitations for crop growth, which can assist managers to prioritize soil management efforts and provide insights into the areas where investments in soil quality would be most effective.

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